

Physics of Complex Systems Homework 9

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Problem 1 (Entanglement of a Pure 2-Qubit State). Consider the 2-qubit state

$$\rho = |\psi\rangle\langle\psi|$$

with

$$|\psi\rangle = \frac{1}{5}|00\rangle + \frac{2i\sqrt{2}}{5}|01\rangle - \frac{4}{15}|10\rangle - \frac{8i\sqrt{2}}{15}|11\rangle.$$

- (a) Use Mathematica or similar tools to compute the 4×4 matrix ρ in the qubit basis. Verify that ρ is pure.
- (b) Compute the reduced density matrices $\rho^{(L)}$ and $\rho^{(R)}$ for the left and right qubit by carrying out the partial traces

$$\rho_{jk}^{(L)} = \sum_{m=0}^1 \rho_{jm,km}, \quad \rho_{mn}^{(R)} = \sum_{j=0}^1 \rho_{jm,jn}.$$

- (c) A pure state is called entangled if the reduced density matrices are mixed. Find out whether ρ is entangled or not.

Problem 2 (Quantum Cryptography). Encrypted communication requires that both parties share a key (a random sequence of classical bits) that can be used to encode and decode messages by bitwise XOR operations. The problem of encryption therefore reduces to safely generating identical keys on both sides.

Alice randomly generates a classical bit $a \in \{0, 1\}$, using it to prepare a qubit

$$|\psi\rangle = \begin{cases} |0\rangle, & a = 0, \\ |+\rangle, & a = 1, \end{cases}$$

where

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$$|\pm\rangle = \frac{1}{\sqrt{2}}(|0\rangle \pm |1\rangle).$$

She sends this qubit to Bob.

Likewise Bob randomly generates a classical bit $b \in \{0,1\}$. He chooses one of the two measurements

$$M_0 = |0\rangle\langle 0| - |1\rangle\langle 1|, \quad M_1 = |+\rangle\langle +| - |-\rangle\langle -|.$$

Bob measures the incoming qubit and communicates the measurement result $m = \pm 1$ publicly. They repeat the protocol until $m = -1$.

- (a) Show that $a = 1 - b$ if $m = -1$.
- (b) How many qubits are on average needed to generate one bit of the key?
- (c) Suppose Eve intercepts the communication and measures using

$$E = |\phi_+\rangle\langle\phi_+| - |\phi_-\rangle\langle\phi_-|,$$

where

$$|\phi_+\rangle = \alpha|0\rangle + \beta|1\rangle,$$

$$|\phi_-\rangle = \beta^*|0\rangle - \alpha^*|1\rangle,$$

with

$$|\alpha|^2 + |\beta|^2 = 1.$$

Show that Eve changes the measurement statistics seen by Alice and Bob, depending effectively only on

$$u := |\alpha|^2, \quad v := \frac{1}{2}|\alpha + \beta|^2.$$

- (d) Determine u and v such that Bob and Alice cannot detect the interception.
- (e) In the case of part (d), is it possible for Eve to obtain information about the key?